DAILY EDITORIAL SUMMARY

05th May 2026

Topics to be covered:

- ✓ **Modern Space Warfare**
- ✓ **Buddhism as India's Soft Power**
- ✓ **Sachet Application and Cell Broadcasting**
- ✓ **Renewable Energy and Energy Security**

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Topics

Modern Space Warfare

GS III: Basics of Cyber Security

Context:

The **2022 Russia-Ukraine conflict** demonstrated that modern warfare begins silently; Russia disabled Ukraine's Viasat satellite network via cyberattack before any physical destruction occurred

About 'Concept of "Bricking":

- This refers to **turning a functional satellite into a "brick"**—an object that remains physically intact but is functionally dead.
- **War Defined by Effect:** War is no longer just physical destruction; it is measured by its impact, such as the total loss of internet, UPI payments, and communication.

Three Silent Weapons of Space War:

- **Jamming**: Using cyber attacks to prevent a satellite from sending or receiving signals, rendering it useless.
- **Spoofing:** Sending false GPS coordinates to misdirect ships or planes, potentially leading to accidents.
- **Ground Station Hacking:** Taking control of the physical facilities that manage satellites to seize command of the assets.

The Legal Blindspot - UN Charter and Use of Force

- The silent nature of space warfare creates a significant legal challenge. Because attacks like **jamming or spoofing do not involve physical destruction** or immediate "gunfire," they exist in a gray area of international law.
- Without **"solid proof"** of who initiated a cyberattack, it is nearly impossible for a country to take a grievance to the **United Nations** or claim a formal violation of the use of force.

The Attribution Gap

- **There is a ma**ssive "attribution gap" in space warfare, meaning it is extremely difficult to pinpoint the origin of an attack.
- Even when a nation's satellite is disabled, there is **strategic anonymity**; the victim cannot prove which country is responsible, making it impossible to retaliate or seek diplomatic recourse.

Why Attackers Remain Invisible?

- **Private Proxies: State actors** often use private hackers rather than direct military units to conduct operations, ensuring the government's name is never directly attached to the attack.
- **Multi-layered IP Spoofing:** Hackers use **multiple layers** of internet routing, bouncing their signals through several different countries (e.g., from Country A to B to C) before hitting the target.
  - They **use spoofed or fake IP addresses** to ensure the attack cannot be traced back to its true origin.

The Dual-Use Dilemma & Collapse of Civilian-Military Divide

- **The 1967** Outer Space Treaty mandates that civilian assets should not be targeted during war. However, the **"dual-use dilemma"** arises because modern satellites provide both civilian and military services simultaneously.
  - **For example,** the **Starlink** project provides civilian internet but has also been used by the Ukrainian military to launch drones.
- This causes a total **collapse of the distinction** between civilian and military targets; if a country targets a satellite to stop military operations, they inevitably destroy essential civilian services as well.

India's Response 2026

- In February 2026**, India hosted the Def-Sat Conference** in New Delhi to address these emerging threats.
- The government acknowledged that India cannot remain a passive spectator and must prepare f**or "silent" space wars** that could disable national infrastructure.

CERT-In Guidelines

- **During the 2026 conference,** CERT-In (**Computer Emergency Response Team India)**, in collaboration with SIA-India, issued specific cybersecurity guidelines for the space industry. Central to these guidelines is the **"Secure by Design" doctrine.**
- This approach mandates that cybersecurity must be integrated into every single stage of a satellite's lifecycle: from initial **production and design** to **launch**, its time in **orbit**, and finally its **decommissioning**.

The Global South Problem

- The vulnerability is **most acute for "Third World" or Global South countrie**s. Unlike India, which has some level of self-reliance, many poorer nations have no satellites of their own and rely entirely on foreign providers.
- This means their entire national systems—including **banking and communications**—can be "destroyed in a flash" if the foreign satellite service they rely on is attacked.

Orbital Dependency

- Modern life ha**s developed a critical "orbital dependency,"** where the loss of satellite function leads to cascading failures.
- Because daily activities like **UPI payments**, internet browsing, and GPS navigation are tied to space assets, "bricking" a satellite (turning it into a useless "box" in space) has the same devastating impact as physical war.

Conclusion:

Space warfare creates **cascading failures** and requires a focus on cyber resilience rather than just physical defense.

Buddhism as India's Soft Power

GS I: Indian Culture Salient aspects of Art Forms, Literature and Architecture from ancient to modern times.

Context:

While Jerusalem is the heart of Christianity and Mecca/Medina the center of Islam, India has not fully leveraged its status as the birthplace of Buddhism to attract global pilgrims and influence.

About India's Buddhist Landscape:

- **Historical Authenticity**: The emergence of Buddhism from the Indian subcontinent establishes India as the authentic birthplace and nurturing ground of Buddhist philosophy and tradition.
- **Sacred Geography of Buddhism in India:**
  - **Bodh Gaya:** Site of Enlightenment.
  - **Sarnath:** Location of the First Sermon (Dharmachakra Pravartana).
  - **Kushinagar:** Site of Mahaparinirvana.
  - **Nalanda:** Historically a global hub for knowledge.

Spread and Evolution of Buddhism:

- The sources detail how Buddhism migrated from India and adapted to local cultures, forming **Theravada** (Sri Lanka), **Vajrayana** (Tibet), **Chan** (China), and **Zen** (Japan).
- **Pan-India Buddhist Footprint:** Beyond Bihar and UP, significant sites exist in **South India** (Amravati, Nagarjunakonda), **West India** (Ajanta-Ellora), **Central India** (Sanchi Stupa), and the **Northeast** (Tawang).

- **India as Civilisational Convergence Point:** India's Buddhist circuit—centred around sites like Bodh Gaya, Sarnath, and Kushinagar—can serve as a **global meeting ground for all Buddhist traditions**.
- **Civilisational Reunion, Not Tourism:** This integration is not merely a **tourism initiative**, but a **civilisational reunion**, where diverse Buddhist traditions symbolically return to their shared origins in India

Economic Potential:

- Buddhist tourism **contributes to 6% of foreign tourist arrivals**; scaling this could significantly **boost the GDP (currently 5.22% from tourism)** and develop impoverished regions like Bihar.

Current Challenges:

- Lack of **integrated circuit planning and poor connectivity** between sites.
- **Weak infrastructure,** including a lack of world-class hotels and multilingual guides.
- **Fragmented governance** between state policies (UP vs. Bihar).
- **Cross-border constraints** with Nepal's Lumbini.

Future Strategy:

- **Establish** a **dedicated authority for Buddhist tourism** for single-window clearance.
  - Develop Bodh Gaya as an international aviation hub and improve high-speed rail links.
  - Revive **Pali scholarship** and global research to regain intellectual leadership.
- **Global Relevance:** Buddhism offers solutions for modern conflicts through **Middle Path** diplomacy, mental health (mindfulness), ethical governance, and climate change resilience **(concept of interdependence).**

Conclusion:

India must view Buddhism as a **Civilizational Reunion** tool and use **Relic Diplomacy** to strengthen global ties.

Mains Practice:

Q. Examine the role of India's Buddhist heritage in enhancing its soft power and Relic Diplomacy in Southeast Asia. What systemic bottlenecks hinder the economic potential of the Buddhist Circuit?

(250 Words, 15 Marks)

Sachet Application and Cell Broadcasting

GS 3: Disaster and Disaster Management

Context:

On a recent Saturday at 11:40 AM, an emergency alert test was conducted on mobile phones across India to test the new "Sachet" system.

What is SACHET?

- SACHET is **India's national emergency alert system based on Cell Broadcast technology. Developed by C-DOT** (Centre for Development of Telematics) under the Department of Telecommunications (DoT).

SACHET — Key Features (No Personal Data Needed)

- No app download or subscription required
- Phone just needs to be ON with **Wireless Emergency Alerts enabled in Settings**
- No personal data collected — zero privacy concerns
- Covers tourists & visitors automatically (works based on location)

Cell Broadcast Technology — Basics

- A **method where a short message is sent simultaneously** to all mobile phones present in a specific geographic area — without using individual phone numbers.

- **History & Origin:** Developed by the **European Telecommunications Standards Institute (ETSI)** in the early 1990s.
- **SMS vs. Cell Broadcasting:** Traditional SMS is a one-to-one system that is often delayed during disasters. **Cell Broadcasting** is a one-to-many system that sends messages to every phone in a tower's range instantly, regardless of network congestion or whether the user's number is known.
- **Key Technical Facts (ITU)**
  - Can simultaneously cover millions of users
  - Can also be targeted to a small geographic area
  - Currently adopted by **30+ countries**

How Cell Broadcast Works — The Mechanism?

- **Normal Phone-Tower Communication:** Every mobile phone constantly communicates with the nearest tower to stay connected. This is a quiet, one-way background process users never notice.

The Alert Mechanism (Step-by-Step)

- **Step 1:** Government/authority detects a hazard & activates the system.
- **Step 2:** A single command is sent to towers in the affected geographic area.
- **Step 3:** Each tower broadcasts the warning message to **every phone in range simultaneously** — one push, millions of alerts.
- **Step 4:** Phone rings with distinct sound & vibration; message stays on screen until user presses
- **Indigenous Technology:** Developed by **C-DOT** under the Department of Telecommunications, the system requires no app download and even overrides silent modes with a loud sound until acknowledged.
- **Mechanics:** It utilizes the constant silent communication between cell towers and mobile phones to push alerts to all visitors and residents in a specific area.

Conclusion:

To be effective, authorities must avoid **"Alert Fatigue"** by using the system strictly for real disasters, preventing the public from ignoring future warnings.

Mains Practice:

Q. Effective early warning systems are the linchpin of proactive disaster management. In the light of the recent testing of the SACHET cell broadcast system in India, analyze how this technological shift from SMS to Cell Broadcasting enhances India's disaster resilience and aligns with global best practices.

(250 Words, 15 Marks)

Renewable Energy and Energy Security

GS III: Infrastructure: Energy

Context:

Under the Paris Agreement, India has updated its **National Determined Contributions (NDCs)** to aim for **60% non-fossil fuel power capacity by 2035**.

Evolving Concept of Energy Security

- Energy security now extends beyond mere availability of energy resources.
- It includes **resilience against external disruptions**, such as supply shocks.
- Domestic capacity to sustain energy supply during crises has become crucial.

India's Updated Renewable Energy Targets

- India initially committed to achieving 50% installed power capacity from non-fossil fuels by 2030.
- The updated NDC raises this ambition to **around 60% non-fossil fuel capacity by 2035**.
- This reflects India's increasing emphasis on **renewable energy transition and decarbonisation**.

Gap Between Installed Capacity and Actual Generation

- India has achieved over 250 GW of non-fossil installed capacity.
- However, this contributes only **about 25% to total electricity generation**.
- This highlights the issue of **underutilisation of renewable infrastructure**.

Intermittency in Renewable Energy

- Renewable sources like solar and wind are **inherently intermittent**.
- Solar power is available only during daytime, while wind energy depends on wind conditions.
- This variability creates challenges in ensuring **continuous power supply**.

Regional Energy Imbalance and Curtailment

- States like **Bihar, Jharkhand, and Chhattisgarh** face rising electricity demand.
- States like **Rajasthan and Gujarat** often experience **curtailment of excess renewable energy**.
- This paradox arises due to **inadequate transmission and last-mile connectivity**.

Need for Energy Storage Solutions

- Excess renewable energy can be stored using **battery storage systems**.
- Storage helps in managing intermittency and ensuring supply during non-generation hours.
- However, India currently faces **limited battery infrastructure and high costs**.

The Paradox of Waste:

- States like Rajasthan and Gujarat often waste (curtail) solar energy because they lack **last-mile connectivity** to transmit it to high-demand areas like Bihar and Jharkhand.

Pricing and Grid Stability:

- The grid faces immense pressure transitioning from solar power during the day to coal at night.
- There is a need for **separate pricing** for "peak demand" availability and "flexibility" to stabilize the grid, similar to models in the UK and USA.

Financial Health of Discoms:

- Power distribution companies are losing trillions due to **cross-subsidization** (charging industries more to provide free power to others).

Proposed Solutions:

- **Feeder-level Segregation:** Separating agricultural electricity lines from residential lines.
- **Solar Micro-grids:** Connecting separated agricultural feeders to solar plants under the **PM-KUSUM** scheme.

Conclusion:

Energy security requires moving beyond "installed capacity" to focus on **battery storage**, grid flexibility, and the financial reform of Discoms.

Mains Practice:

Q. Despite achieving significant milestones in renewable energy capacity addition, India's power grid remains vulnerable to external shocks and internal inefficiencies. Critically analyze the structural challenges in India's power transition and suggest measures for achieving a reliable, round-the-clock clean energy ecosystem. (250 Words, 15 Marks)

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